

The Bernoulli Distribution

2026-04-17

Introduction

The Bernoulli distribution models a single binary trial: success vs. failure, disease vs. healthy, click vs. no click. It is the simplest non-trivial distribution and the building block of the binomial, geometric, and negative binomial distributions.

Prerequisites

Random variables and basic probability.

Theory

A random variable X is **Bernoulli**(p) if it takes values in $\{0, 1\}$ with

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

PMF: $p_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$.

Moments:

- $E[X] = p$.
- $\text{Var}(X) = p(1-p)$, maximised at $p = 0.5$ (value 0.25).
- $E[X^k] = p$ for every $k \geq 1$ (since $X^k = X$).

The sum of n iid Bernoulli(p) variables is **Binomial**(n, p).

MLE of p from an iid sample $X_1, \dots, X_n$: $\hat{p} = \bar{X}$, the sample proportion.

Assumptions

Independence across trials and a common probability p . Violations of either produce overdispersion (if p varies) or correlation (if trials are dependent).

R Implementation

```
set.seed(2026)

p <- 0.3
n <- 10000

# Sample from Bernoulli(p) via rbinom with size = 1
X <- rbinom(n, size = 1, prob = p)

# Sample mean is the MLE of p
c(p_hat = mean(X), p_true = p)

# Sample variance approximates p(1 - p)
```

```

c(sample_var = var(X), theoretical = p * (1 - p))

# 95% Wald confidence interval for p
se_p <- sqrt(mean(X) * (1 - mean(X)) / n)
c(lower = mean(X) - 1.96 * se_p, upper = mean(X) + 1.96 * se_p)

# Wilson (better for small p or small n)
binom::binom.confint(sum(X), n, conf.level = 0.95, methods = "wilson")

```

Output & Results

p_hat	p_true
0.3002	0.3000

sample_var	theoretical
0.2101	0.2100

lower	upper
0.2912	0.3092

method	x	n	mean	lower	upper
Wilson	3002	10000	0.3002	0.2912	0.3094

Empirical mean is $0.3002 \approx p$; empirical variance is $0.2101 \approx p(1 - p) = 0.21$; Wald and Wilson CIs agree in this range.

Interpretation

In applied work, the Bernoulli is the underlying distribution of every binary outcome: adverse event, responder status, screen-positive. Reporting usually aggregates to proportions across n independent Bernoullis – i.e., to the binomial framework.

Practical Tips

- For small samples or extreme p (near 0 or 1), use Wilson or Clopper-Pearson CIs instead of Wald.
- Overdispersion (variance exceeding $p(1 - p)$) in grouped data signals unobserved heterogeneity; consider beta-binomial or mixed-effects logistic regression.
- `rbinom(n, size = 1, prob = p)` is the direct way to simulate Bernoullis; `sample(0:1, n, replace = TRUE, prob = c(1 - p, p))` is equivalent.
- The Bernoulli likelihood is the foundation of logistic regression; MLE of p from regression coefficients requires the logit link.
- Two dependent Bernoullis require a specification of their joint distribution – usually via a copula or a 2x2 joint PMF.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots